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Implementation of a generalized least-squares method for determining calibration curves from data with general uncertainty structures

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Abstract
The determination of a best-fit calibration curve that describes the response of a measuring system to the value of a standard is one of the most widely used procedures in metrology. The mathematical basis for a generalized least-squares solution to this problem is reviewed. Examples of the application of a software implementation of the method are presented to illustrate the treatment of calibration problems with different uncertainty structures for the calibration data, including correlated data. The examples concern the calibration of analysers to measure the composition of natural gas and the calibration of a gas flow dilutor.

1. Introduction

Calibration is one of the most widely employed procedures in metrology. Although there are important differences between the procedures used for the calibration of different types of measuring systems, there are many steps that are common. One such step is the computational method used to establish the mathematical function (the calibration curve) that relates the response or indication of the measuring system to the value realized by a standard. A further step is the subsequent use of the calibration curve, for example, to estimate the stimulus corresponding to a measured value of the response (inverse evaluation). A specific challenge is to undertake these steps in such a way as to give valid uncertainties associated with these estimates following internationally accepted guidelines [1].

A method commonly used for determining a calibration curve is ordinary least-squares regression. The method has the advantage that there are many software implementations available. However, implicit in its use is the assumption that the values realized by the standards in the calibration have negligible associated uncertainties relative to those of the measured responses of the system. In many applications, such as those given here, this assumption is inappropriate. Ignoring the uncertainties associated with the values realized by the standards can yield biased results and unreliable uncertainties associated with estimates of quantities derived from the calibration curve.

This paper reviews the generalized least-squares approach to the calibration problem, which accounts properly for the uncertainty structure of the data used to establish the calibration curve [2–4]. Unlike ordinary least-squares regression, the method permits a valid treatment of calibration problems for which there are uncertainties associated with measured values both of the stimulus and the corresponding response of the measuring system, as well as correlations associated with pairs of such values. The approach, and special cases of it, is referred to in various ways in the mathematical, statistical and measurement literature, including 'orthogonal distance regression' [5, 6], 'generalized distance regression' [7], 'errors in variables' [8,9] and 'generalized Gauss–Markov regression' [3].

The paper is organized as follows. The generalized least-squares problem is reviewed together with some of its special cases (section 2). The validation (section 3) and use (section 4) of a calibration curve established by generalized least-squares regression are discussed, and a software implementation of the method is described (section 5). The performance of the method is illustrated using two examples from analytical chemistry, for which the calibration data have different uncertainty structures. The first concerns the analysis of natural gas (section 6) and the second the calibration of a gas flow dilutor (section 7). A summary is given (section 8).

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2. Determination of a calibration curve

2.1. Calibration data

Calibration involves determining the response v of a measuring instrument or system as a function of a stimulus t, where t is often associated with values realized by standards. Model building involves establishing the functional relationship $v = h(t, \boldsymbol{a})$ between v and t. The model parameters $\boldsymbol{a} = (a_1, \ldots, a_n)^{\mathrm{T}}$ in this relationship parametrize the possible response behaviour over the stimulus interval of concern, and the actual behaviour is approximated in terms of estimates of the parameters determined from the measurement data.

Suppose measurement data

$$\boldsymbol{x} = \begin{pmatrix} \boldsymbol{v} \\ \boldsymbol{t} \end{pmatrix}, \tag{1}$$

comprising measured values $\boldsymbol{v} = (v_1, \ldots, v_m)^{\mathrm{T}}$ of the response corresponding to values $\boldsymbol{t} = (t_1, \ldots, t_m)^{\mathrm{T}}$ of the stimulus are gathered. The uncertainty matrix (covariance matrix)

$$\boldsymbol{U}_{\boldsymbol{x}} = \begin{pmatrix} \boldsymbol{U}_{\boldsymbol{v}} & \boldsymbol{U}_{\boldsymbol{vt}} \\ \boldsymbol{U}_{\boldsymbol{tv}} & \boldsymbol{U}_{\boldsymbol{t}} \end{pmatrix} \tag{2}$$

of order $2m$ containing information about the uncertainties associated with $\boldsymbol{x}$ is assumed to be available. The sub-matrix $\boldsymbol{U}_{\boldsymbol{v}}$ of order m contains the covariance (mutual uncertainty) $u(v_i, v_j)$ associated with the measured values v_i and v_j, with $u(v_i, v_i) = u^2(v_i)$, the variance (squared standard uncertainty) associated with v_i. Similarly, $\boldsymbol{U}_{\boldsymbol{t}}$ and $\boldsymbol{U}_{\boldsymbol{vt}} = \boldsymbol{U}_{\boldsymbol{tv}}$ contain, respectively, the uncertainty information $u(t_i, t_j)$ and $u(v_i, t_j)$.

2.2. Generalized least-squares regression

Let $\boldsymbol{r} = (r_1, \ldots, r_m)^{\mathrm{T}}$ and $\boldsymbol{s} = (s_1, \ldots, s_m)^{\mathrm{T}}$, defining points (s_i, r_i) on a calibration curve defined by model parameters $\boldsymbol{a}$, denote, respectively, the quantities of which $\boldsymbol{v}$ and $\boldsymbol{t}$ are realizations. In generalized least-squares regression, estimates $\hat{\boldsymbol{a}}, \hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{s}}$ are sought such that $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{s}}$ are 'close to' the measured values $\boldsymbol{v}$ and $\boldsymbol{t}$ (in a sense defined below) and are constrained to lie on a calibration curve defined by $\hat{\boldsymbol{a}}$.

Formally, the generalized least-squares problem is to solve [2–4]

$$\min_{\boldsymbol{r},\boldsymbol{s},\boldsymbol{a}} \begin{pmatrix} \boldsymbol{r} - \boldsymbol{v} \\ \boldsymbol{s} - \boldsymbol{t} \end{pmatrix}^{\mathrm{T}} \boldsymbol{U}_{\boldsymbol{x}}^{-1} \begin{pmatrix} \boldsymbol{r} - \boldsymbol{v} \\ \boldsymbol{s} - \boldsymbol{t} \end{pmatrix} \tag{3}$$

subject to the constraints

$$r_i = h(s_i, \boldsymbol{a}), \qquad i = 1, \ldots, m, \tag{4}$$

in which it is assumed that $\boldsymbol{U}_{\boldsymbol{x}}$ has full rank. The importance of generalized least-squares regression comes from the fact that it represents maximum likelihood estimation for measured values constituting realizations of normally distributed quantities. The Gauss–Markov theorem [10] states that, with some mild qualifications, the generalized least-squares approach makes the 'best' use of the available data $\boldsymbol{x}$ and $\boldsymbol{U}_x$.

Denoting by $\boldsymbol{h}(\boldsymbol{s}, \boldsymbol{a})$ the m-vector of elements $h(s_i, \boldsymbol{a})$ and defining

$$\boldsymbol{e}(\boldsymbol{s}, \boldsymbol{a}) = \begin{pmatrix} \boldsymbol{h}(\boldsymbol{s}, \boldsymbol{a}) - \boldsymbol{v} \\ \boldsymbol{s} - \boldsymbol{t} \end{pmatrix}, \tag{5}$$

the minimization (3) and the constraints (4) may be combined to give

$$\min_{\boldsymbol{s},\boldsymbol{a}} \boldsymbol{e}(\boldsymbol{s}, \boldsymbol{a})^{\mathrm{T}} \boldsymbol{U}_{\boldsymbol{x}}^{-1} \boldsymbol{e}(\boldsymbol{s}, \boldsymbol{a}). \tag{6}$$

Furthermore, writing

$$\boldsymbol{L}\boldsymbol{f}(\boldsymbol{s}, \boldsymbol{a}) = \boldsymbol{e}(\boldsymbol{s}, \boldsymbol{a}), \tag{7}$$

where $\boldsymbol{L}$ is the lower triangular matrix of order $2m$ in the Cholesky factorization $\boldsymbol{U}_x = \boldsymbol{L}\boldsymbol{L}^{\mathrm{T}}$ of $\boldsymbol{U}_x$ [11], problem (6) becomes

$$\min_{\boldsymbol{s},\boldsymbol{a}} \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{s}, \boldsymbol{a}) \boldsymbol{f}(\boldsymbol{s}, \boldsymbol{a}) \equiv \min_{\boldsymbol{s},\boldsymbol{a}} \sum_{i=1}^{2m} f_i^2(\boldsymbol{s}, \boldsymbol{a}), \tag{8}$$

which is in the form of a standard non-linear least-squares problem. The uncertainty matrix $\boldsymbol{U}_y$ associated with a solution $\boldsymbol{y} = (\hat{\boldsymbol{s}}^{\mathrm{T}}, \hat{\boldsymbol{a}}^{\mathrm{T}})^{\mathrm{T}}$ to problem (8) is given, formally, by

$$\boldsymbol{U}_{\boldsymbol{y}} = \begin{pmatrix} \boldsymbol{U}_{\hat{\boldsymbol{s}}} & \boldsymbol{U}_{\hat{\boldsymbol{s}}\hat{\boldsymbol{a}}} \\ \boldsymbol{U}_{\hat{\boldsymbol{a}}\hat{\boldsymbol{s}}} & \boldsymbol{U}_{\hat{\boldsymbol{a}}} \end{pmatrix} = (\boldsymbol{J}^{\mathrm{T}}\boldsymbol{J})^{-1}, \tag{9}$$

where $\boldsymbol{J}$ is the Jacobian matrix of order $2m \times (m+n)$ containing the derivatives of $\boldsymbol{f}(\boldsymbol{s}, \boldsymbol{a})$ with respect to $\boldsymbol{s}$ and $\boldsymbol{a}$ evaluated at the solution.

The solution to problem (8) may be obtained by the application of a standard numerical algorithm, for example, the Gauss–Newton algorithm [12]. The Gauss–Newton algorithm in general converges linearly at a rate that depends on the condition of the least-squares problem, the size of the weighted residual deviations $\boldsymbol{f}$ near the solution and the curvature of the deviations. If the problem is well conditioned, the weighted residual deviations are small and nearly linear in the parameters, then the convergence can be expected to be fast [13]. If $\boldsymbol{U}_{\boldsymbol{x}}$ is poorly conditioned, for example, as a consequence of strong correlation effects, the above approach may be numerically unstable. A stable approach to implementing the Gauss–Newton algorithm in this case is to use the generalized QR-factorization [14, 15].

The above description of generalized least-squares regression for solving a calibration problem is completely general in terms of the nature of the uncertainties associated with the calibration data and the form of the calibration curve. In the following sections, some particular instances of generalized least-squares regression that arise in practice are described.

2.3. Weighted least-squares regression

Suppose the measured values t_i are assumed to have negligible associated uncertainty and the v_i are realizations of mutually independent quantities. In this case, $\boldsymbol{x}$ is replaced by $\boldsymbol{v}$ and $\boldsymbol{U}_{\boldsymbol{x}}$ by $\boldsymbol{U}_{\boldsymbol{v}} = \mathrm{diag}(u^2(v_1), \ldots, u^2(v_m))$. With $\boldsymbol{s} = \boldsymbol{t}$ and writing

$$f_i(\boldsymbol{a}) = \frac{1}{u(v_i)}(h(t_i, \boldsymbol{a}) - v_i), \qquad i = 1, \ldots, m, \tag{10}$$

the generalized least-squares problem reduces to the (familiar) weighted least-squares problem

$$\min_{\boldsymbol{a}} \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{a}) \boldsymbol{f}(\boldsymbol{a}) \equiv \min_{\boldsymbol{a}} \sum_{i=1}^{m} \left(\frac{h(t_i, \boldsymbol{a}) - v_i}{u(v_i)} \right)^2. \tag{11}$$

In this way, weighted least-squares regression may be regarded as an instance of generalized least-squares regression in which only the measured data $\boldsymbol{v}$ are subject to uncertainty.

2.4. Distance regression

Suppose the measured data t_i and v_i are realizations of mutually independent quantities, i.e. $\boldsymbol{U}_t = \operatorname{diag}(u^2(t_1), \ldots, u^2(t_m))$, $\boldsymbol{U}_v = \operatorname{diag}(u^2(v_1), \ldots, u^2(v_m))$ and $\boldsymbol{U}_{tv} = \boldsymbol{0}$. Then,

$$\begin{aligned} f_i(\boldsymbol{s}, \boldsymbol{a}) &= \frac{1}{u(v_i)}(h(s_i, \boldsymbol{a}) - v_i), \\ f_{i+m}(\boldsymbol{s}, \boldsymbol{a}) &= \frac{1}{u(t_i)}(s_i - t_i), \qquad i = 1, \ldots, m, \end{aligned} \tag{12}$$

and the generalized least-squares problem reduces to

$$\begin{aligned} \min_{\boldsymbol{s},\boldsymbol{a}} \boldsymbol{f}^{\mathrm{T}}(\boldsymbol{s}, \boldsymbol{a})\boldsymbol{f}(\boldsymbol{s}, \boldsymbol{a}) &\equiv \min_{\boldsymbol{s},\boldsymbol{a}} \sum_{i=1}^{m} d_i^2, \\ d_i^2 &= \left(\frac{h(s_i, \boldsymbol{a}) - v_i}{u(v_i)}\right)^2 + \left(\frac{s_i - t_i}{u(t_i)}\right)^2, \end{aligned} \tag{13}$$

in which, at the solution, d_i corresponds to the 'generalized' distance between the point (t_i, v_i) and the calibration curve. Furthermore, in the case $u(v_i) = u(t_i)$, d_i corresponds to the orthogonal distance.

3. Validation of a calibration curve

In order to derive valid information from a calibration curve, established in the manner described above, the calibration curve must be shown to be consistent with the data (the measured values and the associated variances and covariances).

If the measured values are subject to additive effects characterized by a (joint) probability distribution D, and the calibration curve and data are consistent, the vector $\boldsymbol{e}$ of residual deviations (equation (5)) evaluated at the solution can be expected to be a sample from D. If it is plausible that $\boldsymbol{e}$ is a sample from D then there is no reason to doubt consistency. A subjective approach to undertaking this test is to examine the 'randomness' of the weighted residual deviations $\boldsymbol{f}$ (equation (7)) evaluated at the solution. This can be done most simply by visual inspection. For a small number of measured values it is often quite difficult to show that non-random behaviour is present. For a large number of measured values non-random behaviour is often apparent.

In the case that D is a multivariate normal distribution with covariance matrix $\boldsymbol{U}_x$, the chi-squared test provides an objective approach to testing consistency. Under the normal assumption, the (observed) chi-squared value

$$\chi^2_{\mathrm{obs}} = \boldsymbol{f}^{\mathrm{T}}(\hat{\boldsymbol{s}}, \hat{\boldsymbol{a}})\boldsymbol{f}(\hat{\boldsymbol{s}}, \hat{\boldsymbol{a}}) \tag{14}$$

is a realization of a random variable that is distributed as $\chi^2(p)$ with $p = m - n$ degrees of freedom. In the chi-squared test, the value of χ^2_{obs} is compared with the expectation of a random variable distributed as $\chi^2(p)$. If χ^2_{obs} is significantly larger than p it is unlikely that the normality assumption holds, i.e. that one or more of the measured values are discrepant. A plot of the normalized residual deviations $e_i/u(e_i)$ may help to identify discrepant measured values, for example, as those measured values for which the normalized residual deviation exceeds two in absolute value [16].

4. Use of a calibration curve

A calibration curve is typically used for the purpose of inverse evaluation, in which a measured value v_0 of the response of the measuring system, with an associated uncertainty $u(v_0)$, is used to determine an estimate t_0 of the corresponding stimulus and the associated uncertainty $u(t_0)$. The estimate t_0 is calculated by solving the equation

$$h(t_0, \hat{\boldsymbol{a}}) = v_0. \tag{15}$$

Since this is typically a non-linear equation (the exception is when h defines a straight-line model), a numerical technique, such as bisection, is used to calculate t_0 [17]. The uncertainty associated with t_0 is evaluated by applying the law of propagation of uncertainty [1, 18] to obtain

$$u^2(t_0) = \left(\frac{\partial h}{\partial t}\right)^{-2} (u^2(v_0) + (\nabla_{\boldsymbol{a}} h)\boldsymbol{U}_{\hat{\boldsymbol{a}}}(\nabla_{\boldsymbol{a}} h)^{\mathrm{T}}), \tag{16}$$

where $\nabla_a h$ is the gradient vector of order $1 \times n$ containing the derivatives of $h(t, \boldsymbol{a})$ with respect to $\boldsymbol{a}$, with all derivatives evaluated at $t = t_0$ and $\boldsymbol{a} = \hat{\boldsymbol{a}}$.

5. Software implementation

XGENLINE[1] is a (stand-alone) software package for ordinary and generalized least-squares regression to measurement data using polynomial calibration curves. The software uses a Gauss–Newton algorithm with line search [12, 13] implemented in terms of high quality modules for numerical linear algebra. It uses the solution to the ordinary least-squares regression problem to provide starting estimates for the generalized least-squares problem. The software provides information to assist with the validation of the determined curve, such as plots of weighted residual deviations, as well as the ability to undertake inverse evaluation using that curve. Currently the software supports generalized distance regression in which for each data point (t_i, v_i) the associated uncertainties $u(t_i)$ and $u(v_i)$ and covariance $u(t_i, v_i)$ may be specified, with all other covariances assumed to be zero. A version of the software that permits more general uncertainty structures is under development, as is a version of the software that takes the form of an 'add-in' to Microsoft Excel.

Polynomials provide an important class of calibration curves. They include models that may often be appropriate for a measuring system with simple non-linear behaviour. Although a representation of polynomials as a power series (in terms of monomial basis functions t^j, $j = 0, 1, 2, \ldots$) is natural, such a representation can lead to an ill-conditioned formulation of the calibration problem [13]. For this reason, XGENLINE uses a representation in terms of Chebsyhev polynomials that has much better numerical properties [19].

Other software implementations of generalized least-squares regression are available, such as B_LEAST [20] and AMC Software [21], although the latter only treats the case of a straight-line calibration curve.

[1] Available from http://www.eurometros.org/.

6. Analysis of natural gas

6.1. Description

The price (per unit volume) at which natural gas is traded depends on its calorific value, a measure of the quantity of energy released when the gas is burned. The calorific value is calculated from measured values of the composition of the gas. Since the amount of natural gas traded is extremely large, it is essential that the composition is determined as well as possible with soundly evaluated uncertainties. State-of-the-art analysis uses a gas chromatograph (GC) to determine the amount fraction of each component species of the gas [22,23]. A feature of natural gas analysis is that all components[2] are analysed in the same sample simultaneously but usually with different chromatographic columns and detectors. The composition of the complete mixture is then established by combining the results for each component.

The example considered here concerns the calibration and use of a single GC for measuring the amount fraction of a particular component of natural gas. The GC is calibrated with respect to a suite of gravimetrically prepared primary standard gas mixtures (PSMs) containing different amount fractions of the component. For each PSM an estimate of the amount fraction is reported together with the associated standard uncertainty. A typical value for the relative standard uncertainty is 0.5%, but larger values may be quoted. The amount fractions for the different PSMs are assumed to be mutually independent.

During a calibration, the response (peak area) of the GC to each PSM was measured. A typical value for the relative standard uncertainty associated with an indication of the GC is 0.5%. The response of the GC is assumed to be a linear function of amount fraction.

6.2. Results

In the example considered here, six PSMs, with amount fractions in the range 0.1 mmol mol^{-1} to 1.7 mmol mol^{-1}, were used to calibrate a GC for the measurement of the amount fraction of *n*-hexane in natural gas. Figures 1 and 2 show the weighted residual deviations associated with, respectively, the ordinary and generalized least-squares straight-line fits to the calibration data (table 1). For the evaluation of the former, no account is taken of the uncertainties associated with the PSMs, whereas for the latter all the available information is used.

The values of the chi-squared statistic associated with the fits are, respectively, 7.1 and 2.1. The former is large compared with the expectation of a random variable distributed as $\chi^2(p)$ with $p = 4$ degrees of freedom and, consequently, there is evidence to reject the hypothesis that a straight-line model is consistent with that part of the data used to establish the ordinary least-squares fit. On the other hand, the value of 2.1 is small compared with the expectation and, although consistency of the model with the calibration data is indicated, it is possible that the uncertainties associated with the data are smaller than those given. Due to the paucity of data, it is difficult to make strong judgements about the fits on the basis of the weighted residual deviations presented in figures 1 and 2.

The indication of the GC for a sample of natural gas with an unknown amount fraction of *n*-hexane was 284 088

[2] The component species typically measured in natural gas analysis are methane, carbon dioxide, nitrogen, ethane, propane, *i*-butane, *n*-butane, *neo*-pentane, *i*-pentane, *n*-pentane and *n*-hexane.

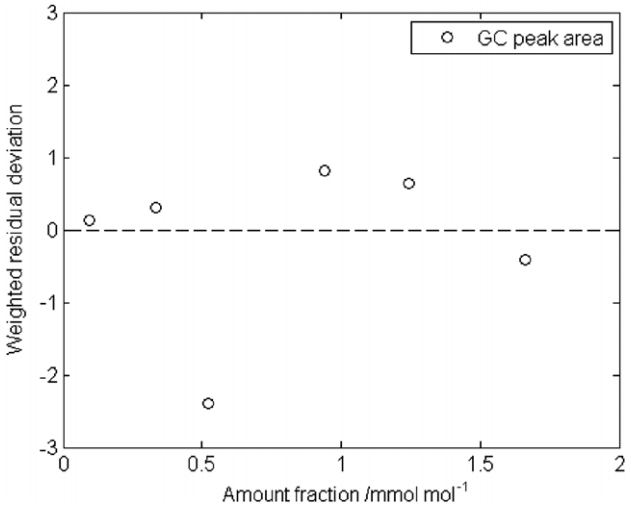


Figure 1. The weighted residual deviations associated with the measured values of GC peak area in the calibration of a GC for measuring the amount fraction of *n*-hexane in natural gas. The calibration curve is a straight-line model determined using ordinary least-squares regression.

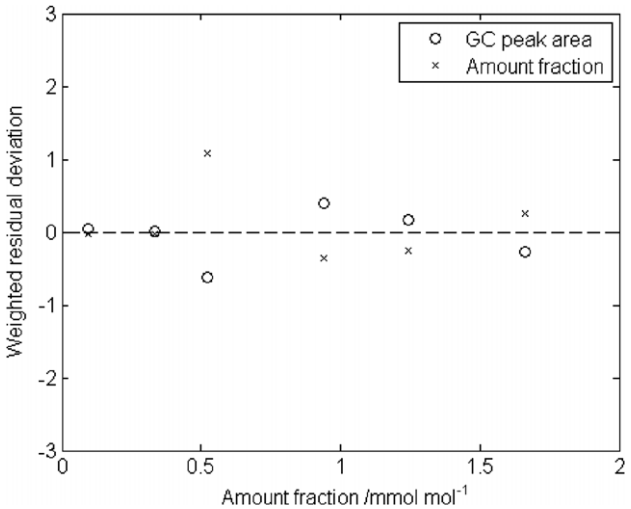


Figure 2. As figure 1, but the calibration curve is determined using generalized least-squares regression, and the weighted residual deviations associated with the measured values of both GC peak area and amount fraction are shown.

Table 1. Data for the calibration of a GC for the measurement of the amount fraction of *n*-hexane in natural gas.

Amount fraction		GC peak area	
Estimate/ mmol mol^{-1}	100 × relative standard uncertainty	Estimate /counts	100 × relative standard uncertainty
0.09	0.5	87 888	0.7
0.33	3.8	310 270	0.5
0.52	1.2	480 730	0.7
0.94	0.4	878 372	0.5
1.24	0.5	1 163 126	0.4
1.66	0.5	1 544 971	0.5

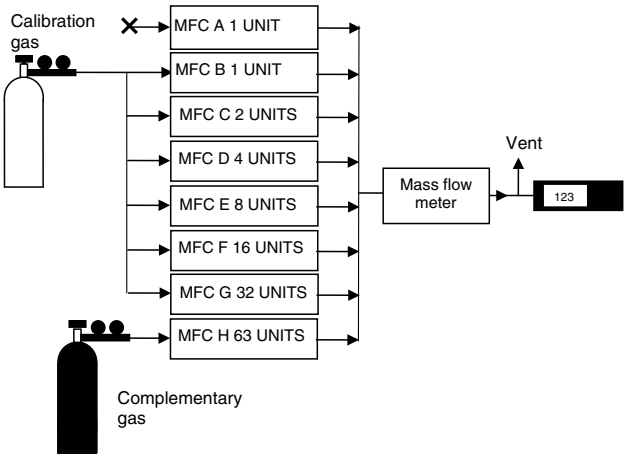


Figure 3. The configuration of binary-weighted gas flow dilutor comprising eight MFCs A to H.

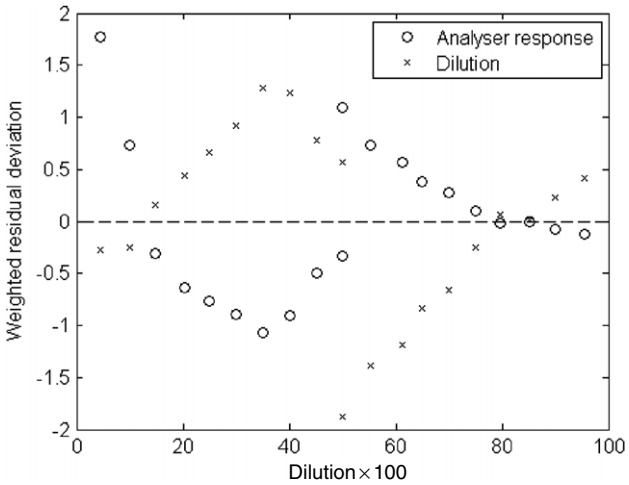


Figure 4. The weighted residual deviations associated with the calibration of an analyser using a binary-weighted gas flow dilutor. The calibration curve is a straight-line model, and the dilutions of the standard gas mixtures used in the calibration are assumed to be obtained independently with relative standard uncertainties of 0.25%.

counts with an associated relative standard uncertainty of 0.5%. On the basis of the generalized least-squares straight-line fit to the calibration data, the estimate of the amount fraction is $0.30\,\mathrm{mmol\,mol^{-1}}$ with an associated relative standard uncertainty of 0.6%. This uncertainty is appreciably smaller than that for the PSM having an amount fraction of *n*-hexane closest to the estimate (3.8% relative) and better (by a factor of two, the square root of the degrees of freedom p) than the 'average' uncertainty for all the PSMs (about 1% relative).

7. Calibration of a gas flow dilutor

7.1. Description

Gas flow dilutors are used to generate adjustable dilutions of a calibration gas by mixing a flow of pure complementary gas (often known as diluent) with a flow of the calibration gas taken from a standard gas mixture at a known dilution (figure 3). In this way, they permit the range of traceable dilutions that can be generated from a single standard gas mixture to be extended. However, it is essential that dilutors are calibrated periodically. This example concerns the calibration of a gas flow dilutor with respect to a reference dilutor, where the latter is a binary-weighted gas flow dilutor [24] and the dilutor being calibrated is based on a series of capillaries.

A characteristic of gas flow dilutors is that they may only be used to generate certain dilutions, and it is generally not possible to match the dilutions generated by one dilutor to those of another. Consequently, the calibration strategy adopted is to use the reference dilutor to calibrate the response of a stable analyser and then to compare the response of the same analyser to the dilutor undergoing calibration. The focus here is with that part of the process concerned with the calibration of the analyser using the reference dilutor.

A feature of the binary-weighted gas flow dilutor used as the reference is that it can be self-calibrated by means of a sequence of internal flow comparisons [24]. This process results in an unusual uncertainty structure associated with the generated dilutions with correlations between pairs of dilutions. The derivation of the uncertainty structure is described in the appendix.

7.2. Results

In the example considered here, a binary-weighted gas flow dilutor was used to generate 20 gas mixtures with dilutions of the calibration gas between 4.5% and 95.5%. During the calibration the response of the analyser to each mixture was measured. The relative standard uncertainty associated with an indication of the analyser was evaluated by averaging data over varying time intervals and was found to be 0.1% of full-scale for a two-minute average.

A simple uncertainty structure is often assumed for the dilutions of the calibration gas in the mixtures, namely the dilutions are independent and the relative standard uncertainties associated with their estimates are equal. Figure 4 shows the weighted residual deviations associated with a generalized least-squares straight-line fit to the calibration data established on the basis of this uncertainty structure and relative standard uncertainties associated with the estimates of the dilutions of 0.25%. There is considerable structure present in the weighted residual deviations. The structure remains in the case that the calibration curve is chosen to be a quadratic polynomial function. For both straight-line and quadratic fits, there is no strong evidence on the basis of the chi-squared test to reject the hypotheses that the fits are consistent with the data.

In the appendix it is indicated how knowledge of the configuration of the reference dilutor, as well as the procedure employed to set up and use the dilutor, provides the basis for obtaining the standard uncertainties associated with estimates of the dilutions as well as the covariances for the dilutions. A consequence is that the dilutions obtained from the same mode of operation of the dilutor are positively correlated, whereas those from different modes are negatively correlated. A further consequence is that the relative standard uncertainty associated with the dilution varies essentially in inverse proportion to the dilution. Figure 5 shows the same information as in figure 4 but for a generalized least-squares straight-fit line fit established on the basis of this more complete uncertainty structure. Taking

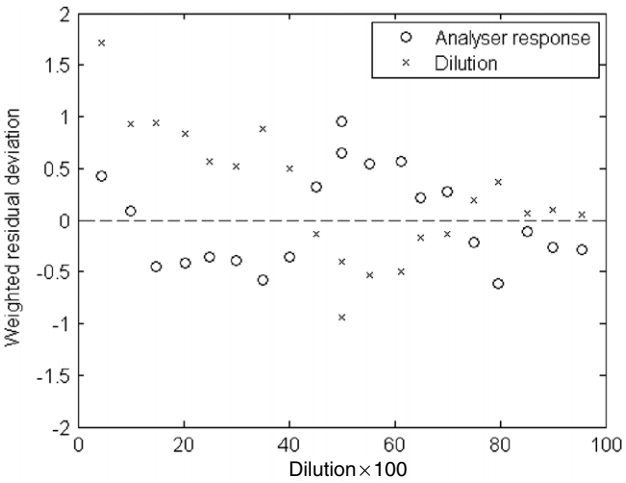


Figure 5. As figure 4, but using an uncertainty structure for the dilutions derived from models that describe the procedure for setting up and using the dilutor (see appendix).

account of correlation effects is beneficial in that the structure of the weighted residual deviations observed in figure 4 is now less evident. Although overall the model is now consistent with the data, further work will be carried out to refine the model in an attempt to accommodate the remaining structure visible in figure 5.

8. Summary

The use of generalized least-squares to establish a calibration curve on the basis of all available information has been reviewed. Consideration has been given to the validation and subsequent use of the calibration curve. Two examples from the area of analytical chemistry have been presented to illustrate calibration problems. They have different uncertainty structures for the calibration data, and from a previously published example [25]. The first example is concerned with the calibration of an instrument to measure the composition of natural gas. The second example is concerned with the calibration of a gas flow dilutor and illustrates a calibration problem in which knowledge of the way it operates is used to establish the uncertainty structure for the calibration data. The calibration curve established on the basis of this uncertainty structure has been shown to be more satisfactory than that established on the basis of simpler assumptions.

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Table 2. The sequence of operations to set up the binary-weighted network with configuration shown in figure 3.

	Operation
1	Activate B; set to give 1 unit flow on the flow meter; close
2	Activate A; set to give an equal 1 unit flow on the flow meter; close
3	Activate A + B; read 2 units flow on the flow meter; close
4	Activate C; set to an equal 2 units flow on the flow meter; close
5	Activate A + B + C; read 4 units flow on the flow meter; close
6	Activate D; set to an equal 4 units flow on the flow meter; close
7	Activate A + B + C + D; read 8 units flow on the flow meter; close
8	Activate E; set to an equal 8 units flow on the flow meter
	Etc

Appendix A. Statistical model for binary-weighted gas flow dilutor

The configuration of the binary-weighted gas flow dilutor used as the reference dilutor is illustrated in figure 3. The dilutor comprises seven mass flow controllers (MFCs), labelled A to H, which may be connected to cylinders of the calibration and complementary gases. The procedure by which the dilutor is used to generate mixtures at different dilutions (concentrations) of the calibration gas comprises two parts. Firstly, setting up the MFCs to generate specific flows of gas (appendix A.1) and, secondly, combining the outputs of the MFCs to generate mixtures at specific dilutions (appendix A.2). Statistical models for the two parts are presented and form the basis for the uncertainty structure associated with the generated dilutions.

Appendix A.1. Setting up the MFCs

The sequence of operations followed to set up the MFCs to generate specific flows of gas is given in table 2.

Let f_B be the flow provided by MFC B, which is set to give a reading on the flow meter of 1 unit of flow, i.e.

$$f_B = 1 + e_B, \tag{A1}$$

where e_B is a sample of a random variable with expectation zero and standard deviation σ that characterizes the repeatability of the flow provided by MFC B when it is activated.

Let f_A be the flow provided by MFC A, which is set to give a reading on the flow meter equal to that obtained when MFC B is activated, i.e.

$$f_A = f_B + e_A = (1 + e_{B,1}) + e_{A,1} + e_A, \tag{A2}$$

where $e_{A,1}$ and $e_{B,1}$ are samples of independent random variables with expectations zero and standard deviations σ. They are used to denote 'instances' of the random variables associated with the activation of MFCs A and B. They correspond to effects that will be common to all (subsequent) activations of MFCs A and B. In contrast, e_A and e_B will be different for each such activation.

Let f_C be the flow provided by MFC C, which is set to give a reading on the flow meter equal to that obtained when MFCs A and B are activated together, i.e.

$$\begin{aligned} f_C &= f_A + f_B + e_C \\ &= (1 + e_{B,1} + e_{A,1} + e_{A,2}) + (1 + e_{B,2}) + e_C \\ &= (2 + e_{A,1} + e_{A,2} + e_{B,1} + e_{B,2}) + e_C, \end{aligned} \tag{A3}$$

where $e_{A,2}$ and $e_{B,2}$ are further instances of the random variables associated with the activation of MFCs A and B, and e_C is a sample of a random variable with expectation zero and standard deviation proportional to the flow provided by MFC C, i.e. 2σ. Since the random variables are independent, it follows from model (A3) that the standard uncertainty $u(f_C)$ associated with the flow provided by MFC C is given by

$$\begin{aligned} u^2(f_C) &= u^2(e_{A,1}) + u^2(e_{A,2}) + u^2(e_{B,1}) + u^2(e_{B,2}) + u^2(e_C) \\ &= \sigma^2 + \sigma^2 + \sigma^2 + \sigma^2 + (2\sigma)^2 = 8\sigma^2. \end{aligned} \tag{A4}$$

Let f_D be the flow provided by MFC D, which is set to give a reading on the flow meter equal to that obtained when MFCs A, B and C are activated together, i.e.

$$\begin{aligned} f_D &= f_A + f_B + f_C + e_D \\ &= (4 + 2e_{A,1} + e_{A,2} + 2e_{B,1} + e_{B,2} + e_{A,3} + e_{B,3} + e_{C,1}) + e_D, \end{aligned} \tag{A5}$$

where e_D is a sample of a random variable with expectation zero and standard deviation proportional to the flow provided by MFC D, i.e. 4σ. It follows from model (A5) that the standard uncertainty $u(f_D)$ associated with the flow provided by MFC D is given by

$$\begin{aligned} u^2(f_D) &= (2\sigma)^2 + \sigma^2 + (2\sigma)^2 + \sigma^2 + \sigma^2 + \sigma^2 + (2\sigma)^2 + (4\sigma)^2 \\ &= 32\sigma^2. \end{aligned} \tag{A6}$$

Furthermore, since there are effects common to the flows provided by MFC C and MFC D (represented by the terms $e_{A,1}$, $e_{A,2}$, $e_{B,1}$ and $e_{B,2}$), it follows that the flows are correlated, with covariance given by

$$\begin{aligned} u(f_C, f_D) &= u(e_{A,1}, 2e_{A,1}) + u(e_{A,2}, e_{A,2}) \\ &\quad + u(e_{B,1}, 2e_{B,1}) + u(e_{B,2}, e_{B,2}) \\ &= 2\sigma^2 + \sigma^2 + 2\sigma^2 + \sigma^2 = 6\sigma^2 \end{aligned} \tag{A7}$$

The flows provided by MFC E (8 units), MFC F (16 units) and MFC G (32 units) and MFC H (63 units) are set similarly[3]. The standard uncertainties associated with the flows generated when the MFCs are activated, as well as the covariances for pairs of flows, are evaluated by applying the law of propagation of uncertainty [1, 18] to the models (A1), (A3) and (A5) together with models for the flows provided by MFCs E to H. The standard uncertainties are listed in table 3. The results indicate that the sequence of relative standard uncertainties is close to the series 7, 28, 112, 448 and 1792 required for them to be completely constant and independent of the flow.

[3] The flow provided by MFC H is set to 63 units, not to the next flow in the binary sequence (64 units). This is achieved by calibrating against the flows of MFCs B to G activated together (i.e. excluding MFC A).

Table 3. Estimates of flow and associated standard uncertainties generated by the MFCs B to H of the reference gas flow dilutor (figure 3). The result for MFC A is not shown because it is only used in the set-up procedure and not in the subsequent use of the dilutor[a].

MFC	Flow f/units of flow	$u(f)$/units of flow	$u(f)/f/1$
B	1	σ	σ
C	2	$\sigma\sqrt{8}$	1.41σ
D	4	$\sigma\sqrt{32}$	1.41σ
E	8	$\sigma\sqrt{132}$	1.44σ
F	16	$\sigma\sqrt{548}$	1.46σ
G	32	$\sigma\sqrt{2376}$	1.49σ
H	63	$\sigma\sqrt{9444}$	1.52σ

[a] The data given here correct errors in table 2 of [18].

Appendix A.2. Generating mixtures with different dilutions

In its first mode of operation, a cylinder of the calibration gas is connected to MFCs B to G and a cylinder of the complementary gas to MFC H (figure 3). By either activating or not the different MFCs (B to G) a flow of calibration gas is produced, given by

$$f_1^{\text{gas}} = b_G f_G + b_F f_F + \cdots + b_B f_B, \tag{A8}$$

where each element of the row vector $\boldsymbol{b} = (b_G, \ldots, b_B)$ takes a value zero (not activated) or one (activated). The flow of complementary gas is that provided by MFC H which is always activated, i.e.

$$f_1^{\text{dil}} = f_H. \tag{A9}$$

The dilution x_1 of the calibration gas in the mixture is given by the flow of the calibration gas as a proportion of the total flow, i.e.

$$x_1 = \frac{f_1^{\text{gas}}}{f_1^{\text{gas}} + f_1^{\text{dil}}} = \frac{b_G f_G + \cdots + b_B f_B}{b_G f_G + \cdots + b_B f_B + f_H}. \tag{A10}$$

For example, setting $\boldsymbol{b} = (0, 0, 0, 0, 0, 1)$ gives a dilution of $1/(1 + 63) \approx 1.6\%$, and setting $\boldsymbol{b} = (1, 1, 1, 1, 1, 1)$ gives a dilution of $63/(63 + 63) = 50\%$.

In its second mode of operation, the cylinders of calibration and complementary gas are interchanged in the configuration shown in figure 3. Then, the flows of the calibration and complementary gases are given by

$$f_2^{\text{gas}} = f_{H,} \qquad f_2^{\text{dil}} = b_G f_G + b_F f_F + \cdots + b_B f_B, \tag{A11}$$

where (again) $\boldsymbol{b}$ is a vector of zeros and ones, and the dilution x_2 by

$$x_2 = \frac{f_2^{\text{gas}}}{f_2^{\text{gas}} + f_2^{\text{dil}}} = \frac{f_H}{f_H + b_G f_G + \cdots + b_B f_B}. \tag{A12}$$

This time, setting $\boldsymbol{b} = (0, 0, 0, 0, 0, 1)$ gives a dilution of $63/(63 + 1) \approx 98.4\%$, and setting $\boldsymbol{b} = (1, 1, 1, 1, 1, 1)$ gives a dilution of $63/(63 + 63) = 50\%$.

By combining the two modes of operation, mixtures with a range of dilutions between about 1.6% and 98.4% may be produced. Pairs of dilutions are correlated because they depend on the flows provided by MFCs that are in common. The standard uncertainties associated with the dilutions, as well as the covariances for pairs of dilutions, are evaluated by applying the law of propagation of uncertainty [1, 18] to the models (A10)

and (A12) combined with those for the flows provided by the individual MFCs ((A1), (A3), (A5), etc). A consequence is that dilutions obtained from the same mode of operation of the dilutor are positively correlated, whereas those from different modes are negatively correlated. A further consequence is that the relative standard uncertainty associated with the dilution varies essentially in inverse proportion to the dilution.

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